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Quality Improvement Spillovers: Evidence from the Hospital Readmissions Reduction Program

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Abstract. Quality knowledge spillovers can enhance the overall effectiveness of quality improvement initiatives. We study the presence and moderators of such spillovers in a multi-task service setting, specifically in hospital inpatient care. Leveraging a national quality improvement regulation, the Hospital Readmissions Reduction Program (HRRP), which offers partial incentives for hospitals to reduce readmissions, we employ difference-in-differences econometric models on a nationwide database and find positive quality spillovers in the healthcare sector. Our findings indicate that the implementation of HRRP led to a significant decrease in 30-day readmissions among patients with clinical conditions or insurance types that were not targeted by the policy. Additionally, we find that task similarity played a positive role in promoting quality spillovers, while a hospital's operational focus on target patients (i.e., the proportion of hospital volume targeted by the policy) did not moderate these spillovers. Notably, we observe that hospitals achieved these quality improvements without increasing the intensity of care provided, and that meaningful improvements in quality were associated with up to a 3% reduction in hospitalization costs. This paper contributes novel insights into how regulators and policymakers can design narrow public policies and regulations that achieve broader results by exploiting the beneficial quality improvement spillovers of partial incentives.

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1. Introduction

Financial incentives are commonly used in both the private and public sectors to influence individuals and organizations to improve quality performance or cost. These incentives are prevalent among manufacturers to drive supplier quality improvements and cost reductions (Handfield et al. 2000), regulators, whether in education, transportation, or healthcare (Stecher et al. 2012), and employers to make promotion and bonus compensation decisions (Deméré et al. 2016, Cai et al. 2023). The effectiveness of incentive designs has been extensively studied, yet less attention has been paid to the potential spillover or externality generated by “partial incentives,” where only a subset of tasks or task dimensions are incentivized. In this paper, we study such spillover effects of a penalty incentive program used by regulators to reduce hospital readmissions.

In a seminal paper, Holmstrom and Milgrom (1991) analyze a principal-agent model in which the principal

assigns either multiple tasks for the agent to perform (i.e., multitask setting) or a single task with multiple dimensions (i.e., multidimensional setting). The authors argue that the impact of partial incentives in a multi-task (multidimensional) setting changes depending on whether the tasks are substitutes or complements. The bulk of their paper, however, assumes “all tasks to be equal substitutes in the agent's cost function,” excluding the possibility that some tasks may be complementary. Under this assumption, they show that partial incentives cause the agent to allocate their finite effort toward the incentivized task (task dimension) and away from other tasks (task dimensions). In the presence of substitutability, prior empirical research offers support for this finding. For example, Lu (2012) demonstrates that a new law requiring nursing homes to publicly report limited quality metrics led to an improvement in the reported metrics but a degradation of nonreported metrics. Similarly, Cai et al. (2023) show that, in a manufacturing

setting, providing incentives for a standard task generated negative spillovers to a secondary task.

In contrast, however, many studies find that partial incentives either have no impact on nonincentivized tasks or task dimensions (see Jenkins et al. (1998) and Bonner and Sprinkle (2002) for a review of this literature) or generate positive spillover effects (Muralidharan and Sundararaman 2011, Christ et al. 2012, Harris et al. 2018, Wang et al. 2022). One possible cause for these different results, which Holmstrom and Milgrom (1991) mention but do not fully explore, is that if there are complementarities between tasks in the agent's private cost function, then partial incentives might lead to increased effort in both incentivized and nonincentivized tasks, that is, beneficial spillovers. Further, Hecht et al. (2012) argue that the discrepancy between the conclusions of Holmstrom and Milgrom (1991) and the above-mentioned literature can be explained by misalignments in other underlying assumptions. Specifically, Hecht et al. (2012) underline the sequential task performance assumption in Holmstrom and Milgrom (1991) as a critical assumption that contradicts the possibility of simultaneous task performance in most empirical settings and identify temporal proximity as a moderator for positive spillovers. Hecht et al. (2012) encourage researchers to identify other moderators for spillover effects in multitask settings. This paper joins a small number of recent studies (Jung and Shin 2022, Gallani 2023) that respond to this call to provide a better understanding of spillover effects of partial incentives in multitask settings.

To design financial incentives in the presence of spillovers, firms and regulators need to understand why spillovers occur and the mechanisms behind them. An organization may reach its target quality requirements through the accumulated experience of repeating the same task (i.e., autonomous learning) and through creating new knowledge by deliberately undertaking learning efforts (i.e., induced learning) (Li and Rajagopalan 1998, Lapré et al. 2000, Forker et al. 2020, Krishnan and Ramasubramanian 2023). The impact of a learning process in an organization, however, is not confined to a specific unit (i.e., individual, group) within the organization. Rather, shared resources within the organization, and communication between different units in the organization, lead to knowledge transfer, also known as vicarious learning or knowledge spillover (Argote et al. 2021).

Several recent empirical studies have examined knowledge spillover in various domains, such as corporate and technology innovation (Kim and Valentine 2021, Kong et al. 2022), organizational collaboration (Mohnen 2022), and supply chain management (Chen et al. 2023). Specific to *quality* spillovers, Muthulingam and Agrawal (2016) and Agrawal et al. (2020) have studied quality knowledge spillovers in the manufacturing setting. These studies have found that quality knowledge is “sticky,” it does not readily spill over, and

spilled-over quality knowledge is not sustained over time because of forgetting. In contrast, in the healthcare setting, Deore et al. (2023, 2024) have found that quality knowledge spills over in multidimensional and multitask settings, respectively.

In this paper, we study several novel features of quality improvement spillovers that arise from partial incentives in a service setting. Specifically, we study the Hospital Readmissions Reduction Program (HRRP), initiated by the U.S. Congress as part of the Patient Protection and Affordable Care Act (ACA) of 2010 (dubbed “Obamacare”) to incentivize hospitals and healthcare systems to improve clinical quality and reduce national healthcare costs, specifically those related to unnecessary hospital readmissions.

HRRP imposes financial penalties on hospitals with above-benchmark readmission rates for Medicare patients over age 65 with one of three clinical conditions: acute myocardial infarction (AMI), heart failure (HF), or pneumonia (PN) (additional conditions were added in later years). The benchmark for each hospital is a function of the hospital's case mix and the performance of peer hospitals and is recalculated annually. The performance of each hospital is assessed every year based on its readmissions over a three-year period. Hospitals with readmission rates exceeding the risk-adjusted benchmark are subject to a reduction in *total* Medicare reimbursement of up to 1%, 2%, and 3% in fiscal years 2013, 2014, and 2015 and onward, respectively (The Patient Protection and Affordable Care Act 2010). In the first five years since the inception of HRRP, an average of 73% of hospitals faced penalties each year, with an average Medicare payment adjustment of 0.6%. These penalties amount to about \$2 billion, ranging from \$290 million in 2013 to \$528 million in 2017 (Boccuti and Casillas 2017).

The program has largely been viewed as a success (Zuckerman et al. 2016, Wasfy et al. 2017). However, as mentioned above, HRRP focuses rather narrowly on the 30-day readmission rate of Medicare patients aged 65 and older with one of three clinical conditions (AMI, HF, and PN). The HRRP implementation is especially suitable for exploring quality improvement spillovers because its initial target patients account for only about 5% of total U.S. hospitalizations. With financial incentives tied to such a narrow target, it raises the question of whether improvements for the target patients have beneficially spilled over to those outside the target, or to the contrary, have come at the expense of nontarget patients.

This paper contributes to the literature on quality knowledge spillover of partial incentives in multitask settings by exploring the impact of both task and organizational (agent) features on quality improvement spillovers. Specifically, this paper answers the following questions. (1) Does quality knowledge spill over at hospitals? (2) What are task-level (i.e., patient-level)

contributing factors to spillover effects? (3) What are organizational-level (i.e., hospital-level) contributing factors to spillover effects?

To answer these questions in the context of HRRP, we use difference-in-differences (hereafter DD) models on a national-level database. In brief, we find that (1) quality improvement spillovers occur in the multitask setting of hospitals for patients who share common features with the target patients, (2) clinical condition similarity (a task-level feature) increases the magnitude of spillovers, and (3) operational focus at hospitals (an organizational-level feature) is not a moderator of quality improvement spillovers. In line with the concept of complementarity, our results indicate that two groups of patients that are not directly targeted by the policy benefit from HRRP-driven readmission-reduction efforts: patients with target insurance but nontarget conditions (spillover across condition) and patients with target conditions but nontarget insurance (spillover across insurance). We also find that the spillover effect across condition is stronger when the condition is more clinically similar to target conditions, whereas spillover effects are not affected by the proportion of target patients in a hospital. Finally, we find that hospitals gain quality improvement for target and nontarget patients without intensifying care provided. This finding provides insight into how hospitals have implemented readmission-reduction interventions.

This paper makes the following contributions. First, we show that in a multitask setting, such as hospitals, quality knowledge readily spills over to nontarget tasks (i.e., patients) that have at least one common attribute (clinical condition or insurance) with target tasks. Second, we find clinical condition similarity, a task-level factor, as a moderator for quality knowledge spillover. These findings suggest that not only is there complementarity between incentivized and nonincentivized tasks but also that spillover effects are a function of the degree of complementarity. To this end, the present paper adds to the emerging literature investigating quality improvement spillovers within a service setting (Deore et al. 2023, 2024; Gallani 2023).

Together, this study shows that in service settings, the effects of a quality improvement policy can extend well beyond the tasks specifically targeted by the policy. Spillover effects, however, are not consistent across tasks. The existence of quality improvement spillovers and their moderators offers insights for regulators and policymakers to fully benefit from these effects when designing partial incentive public policy.

2. Background and Hypotheses Development

To aid in the exposition of the rest of the paper, we define four subgroups or cohorts of patient visits

Figure 1. Patient Cohorts

		CONDITION	
		Target	Non-target
INSURANCE	Medicare	A	B
	Private	C	D

(Figure 1). HRRP evaluates a hospital’s performance based on the readmission rates for Medicare patients over the age of 65 with one of three specific clinical conditions: AMI, HF, or PN. We refer to these conditions collectively as the “target conditions” and to Medicare patients aged 65 and older with target conditions as “target patients.” We refer to target patients as Cohort A. Cohort B includes patients with Medicare as their primary insurance but who do not have one of the three target conditions. Cohort C includes patients with one of the three target conditions but with private insurance. Our analysis is centered around identifying the effects of HRRP on these two cohorts of nontarget patients (Cohorts B and C). Lastly, Cohort D includes patients who do not have any of the target conditions and who have private insurance as their primary insurance. We provide cohort summary statistics in Section 3.

The primary goal of HRRP is to incentivize hospitals to achieve lower 30-day readmission rates for target patients (Cohort A). Prior work, primarily in the medical and health economics domains, has shown this to generally be the case in various state and national-level data sets (Desai et al. 2016, Mellor et al. 2017, Wasfy et al. 2017, Demiralp et al. 2018). We expect this to hold true for our national data set, and for completeness, we confirm this result with our data in Section 5. These consistent results, observed across various states and at the national level, coupled with evidence confirming that these improvements are not achieved through hospitals gaming the 30-day readmission metric (Carey and Lin 2015, Zuckerman et al. 2016, Chen and Savva 2018, Chen and Grabowski 2019) or compromising other quality measures such as continuity of care (Deore et al. 2023), indicate that, overall, hospitals have effectively implemented quality improvement initiatives that reduced readmissions for target patients. Our focus in this paper is on how these improvements spill over to nontarget patients.

To design and implement quality improvement initiatives in a hospital, the corresponding caregivers must develop quality knowledge associated with the care delivery process of target patients. That is, to come up with “a set of interrelated work activities repeatedly utilized by individuals or groups that a body of knowledge

demonstrates will yield an optimal result—good patient outcome” (Tucker et al. 2007, p. 894). Acquiring quality knowledge involves two phases of learning that Lapré et al. (2000) refer to as “conceptual” and “operational” learning, both of which require effort on the part of the caregivers. In the conceptual learning phase, nurses and physicians with different specialties coordinate in work groups to identify causes of readmission for target patients (i.e., the acquisition of “know-why” knowledge, that is, why readmissions occur). In the operational learning phase, based on the know-why knowledge, caregivers design and explore solutions to reducing readmissions for target patients and operationalizing them (i.e., the acquisition of “know-how” knowledge). These interventions could include *clinical* care improvement initiatives, such as an increased number of procedures ordered, a longer length of stay (LOS), and enhanced adherence to medication guidelines (Anderson et al. 2012, Andritsos and Tang 2014, Carey 2015, Krishnan and Ramasubramanian 2023), or *nonclinical* care (i.e., care management) improvement initiatives, such as improving patient education for postdischarge care, and follow-up home visits (Bradley et al. 2012, Kripalani et al. 2014). We refer the reader to Hansen et al. (2011) and Brewster et al. (2016) for surveys of the literature on readmission-reduction interventions.

The acquired know-why and know-how knowledge is embedded in three key elements of hospitals: members, tools, and tasks (Argote and Ingram 2000). Members are human resources in hospitals, including physicians and nurses. Tools are nonhuman resources in hospitals, including software and hardware technologies. Tasks are standard routines and procedures in hospitals. New know-how knowledge generally becomes embedded in tools in the form of adopting new technologies (e.g., medical and surgical equipment, information technology support systems) and in tasks in the form of modified procedures (e.g., aspirin administration upon arrival and discharge for AMI patients).

The development of this knowledge is initiated with the primary goal of improving quality outcomes for target patients. However, to the extent that services to patients are complementary and caregivers and resources are shared between target and nontarget patients, accrued knowledge may also benefit other patients treated in the hospital. This is because hospitals are often organized by clinical specialty (e.g., cardiology, internal medicine) and make use of condition-specific “care paths” (Vera and Kuntz 2007). The care path for a condition can encompass multiple departments and each department is responsible for the implementation of quality assurance programs within its operation (rather than the whole process of care) (Vanberkel et al. 2012). Therefore, an improved quality outcome for a patient requires cooperation between multiple caregivers, and indeed, the success of a quality improvement project is a function of the level of

interdisciplinary collaboration in the work group (Nembhard and Tucker 2011). For example, it is estimated that more than 20 healthcare professionals (e.g., physicians with different specialties, nurses) are involved in providing care for a single patient visit in a hospital (Bohmer and Knoop 2007). A caregiver, however, is often a member of more than one work group and is involved in the care delivery process of patients with different clinical conditions (Nembhard and Tucker 2011). Thus, a caregiver may transfer the knowledge acquired in one work group to other work groups, either explicitly or implicitly (Argote and Ingram 2000, Lee et al. 2015).

In addition to organizational members (i.e., caregivers), tools and tasks along the clinical care paths for target patients that are shared with other patients may also contribute to knowledge spillover. For example, consider the case in which a work group identifies hospital-acquired infections as a cause of readmission for target patients, and proposes modifying or eliminating the procedures that may potentially increase the risk of these infections (e.g., blood transfusion). In this case, the quality knowledge embedded in these procedures may impact nontarget patients who undergo the same procedures. This is especially likely for patients with the same clinical condition but different insurance types that go through the same clinical care path (but not necessarily the same nonclinical or postdischarge activities), and thus share all the resources. As a result, quality knowledge developed for target patients is explicitly transferred to the care delivery process of patients with target conditions but private insurance. For example, Andritsos and Tang (2014) show that adherence to medical guidelines, such as aspirin administration and beta blocker administration at arrival and discharge (for AMI), evaluation of left ventricular systolic dysfunction (for HF), discharge instructions (for HF), angiotensin converting system inhibitor or angiotensin receptor blocker (for AMI and HF), and smoking cessation advice (for AMI and HF) improve quality outcomes. Because medical guidelines are developed for specific clinical conditions (regardless of insurance), an increase in a hospital’s adherence to clinical guidelines is likely not limited to patients with a specific insurance type. In contrast, all patients with the same clinical condition will potentially benefit from this improvement.

As discussed in Section 1, following the principle-agent framework of Holmstrom and Milgrom (1991), it is possible, however, that negative spillovers could occur when tasks are substitutes, that is, nontarget patients suffer an increase in readmissions as a result of HRRP. Specifically, HRRP provides partial incentives in that there are quality improvement incentives only for target patients, which might cause the limited resources to be allocated away from nontarget patients. For example, if hospitals initiate a resource-intensive intervention, such as postdischarge home visits (Hansen et al. 2011),

hospitals may focus this nonclinical intervention on target patients and draw resources away from nontarget patients, leading to negative spillovers. Returning to the idea of whether tasks performed for different patient cohorts are compliments or substitutes, such resource-intensive interventions are likely substitutes because there is little return to learning or scale that occurs between patient cohorts. It costs the same amount of money to send a nurse to the home of a target patient or nontarget patient. Thus, if there are incentives for providing such services to target patients, these efforts may well lead to diverting effort and resources away from nontarget patients. In contrast, low-resource interventions, such as improved discharge instructions, are likely complements because the learning effort exerted on behalf of incentivized target patients can easily and cheaply be extended to nontargeted patients. On the whole, given the high priority in healthcare to not harm any patient, we believe that positive spillovers are more likely.

Altogether, we expect quality knowledge to spill over within the hospital and be utilized for nontarget patients. Knowledge spillover may be identified through quality performance measures (Argote and Fahrenkopf 2016), and thus for the specific case of HRRP, we quantify it with the changes in 30-day readmissions for patients with nontarget conditions or nontarget insurance.

Hypothesis 1. *HRRP leads to reduced 30-day readmission rate of nontarget patients, that is, Cohort B (spillover across insurance) and Cohort C (spillover across condition).*

In the next two hypotheses, we consider variations in spillover effects with respect to the clinical condition and operational focus at a hospital (Research Questions 2 and 3).

Although HRRP has only three target clinical conditions, there are dozens of nontarget clinical conditions, and thus there may be heterogeneity in spillover effects across conditions depending on the clinical similarity of a nontarget condition to target conditions. For example, the international classification of diseases system (ICD) classifies both AMI and HF as cardiology conditions, and they are commonly treated in a dedicated cardiology department of the hospital (Andritsos and Tang 2014). Thus, similar to the concept of *specialty ailments* in Deore et al. (2024), we consider nontarget cardiology conditions (i.e., cardiology conditions other than AMI and HF, such as heart valve disorders and coronary atherosclerosis) to be *similar* nontarget conditions, and all other conditions (i.e., all noncardiology conditions) to be *dissimilar*. There is typically more similarity in the care path for patients with conditions similar to each other than for those with dissimilar conditions. As a result, there are more members, tools, and tasks shared between patients with similar conditions, and thus the likelihood of knowledge spillover between

them increases (Schilling et al. 2003, Muthulingam and Agrawal 2016, Deore et al. 2024). For example, caregivers in a cardiology department may store tacit knowledge developed for AMI- and HF-specific interventions, and this knowledge may also affect quality outcomes for patients with heart valve disorders because they are treated in the same department of the hospital (Argote 2012). In addition, the knowledge embedded in tools and tasks may directly impact these patients. Thus, there is likely a higher level of task complementarity between target patients and patients with similar conditions. In contrast, patients with chronic renal failure (a noncardiac condition), for example, may only benefit from these interventions if there is an overlap between work groups and the accrued quality knowledge is intentionally transferred to these patients. Consequently, we expect to observe a stronger spillover effect across condition for patients with conditions that are more similar to target conditions.

Hypothesis 2. *HRRP leads to a larger reduction in 30-day readmission rate of patients with similar nontarget conditions than of patients with dissimilar nontarget conditions.*

In Hypothesis 1, we consider quality improvement spillovers to nontarget patients over all hospitals combined. However, the magnitude of these spillovers may be different depending on the proportion of a hospital's patients that are target patients. Following the literature, we refer to this as the operational focus on target patients, or simply *focus* (Ding 2014, Freeman et al. 2021), although it is sometimes also referred to as *volume* (Deore et al. 2024).

There are two reasons why quality spillover might vary with focus. First, if a hospital has a large proportion of target patients (i.e., high focus), it likely has a larger number of caregivers interacting with target patients, and these caregivers can be conduits of knowledge spillover as they also work with nontarget patients (Deore et al. 2024). A second reason is due to financial incentives. Because hospitals generally get paid for each patient admission, be it an initial admission or a readmission, a reduction in readmissions results in reduced revenue. HRRP is designed to combat this incentive, but only for target patients. If hospitals are revenue maximizing, then they have incentive to reduce readmission-reduction spillovers to nontarget Cohorts B and C. The larger the proportion of patients that are nontarget patients (i.e., the lower the focus on target patients), the stronger the incentive to prevent readmission-reduction spillovers to those nontarget patients. Together, these reasons suggest that hospitals with high focus will exhibit high spillovers.

Hypothesis 3. *HRRP leads to a larger reduction in 30-day readmission rate of nontarget patients (i.e., Cohorts B and C) in hospitals with high operational focus on target*

patients (i.e., Cohort A) than in hospitals with low operational focus on target patients.

3. Data Description

Our study is based on four years of data (2011–2014) from the Nationwide Readmissions Database (NRD) created by the Agency for Healthcare Research and Quality (AHRQ). A key benefit of this database is its size: Each year includes about 15 million observations that comprise approximately half of all hospitalizations in the United States, and the sample is obtained from more than 2,000 hospitals with wide geographical coverage within the country (Healthcare Cost and Utilization Project 2016). These data are particularly well suited to our study because they include detailed information on each hospitalization, such as the diagnosis codes, admission and discharge timing, and hospital characteristics. The database also includes information on patients, including demographics and the expected primary payer (e.g., Medicare, Medicaid, private insurance).

To define a 30-day readmission incident, we follow prior work as well as the definition of 30-day readmissions in HRRP, wherein only the immediately subsequent hospitalization for the patient (if there is one) is considered for the purpose of determining the timing of a readmission. Note that this measure does not take into account how many readmissions occurred within 30 days of discharge from an index hospitalization (Centers for Medicare & Medicaid Services 2016).

The NRD for each year includes hospitalizations based on the discharge date (i.e., a hospitalization that starts in December of year t and ends in January of year $t + 1$ is only included in the $t + 1$ data set). Because the disguised patient identifier codes are reset each year, individual patients cannot be tracked from one year to the next. Further, the data are disguised such that we observe the month of each discharge but not the day of the month of each admission and discharge. Because of these two limitations, to avoid hospitalizations with possibly censored readmissions, we exclude hospitalizations that occur in November or December of each year to allow for one month of possible hospital stay (99% of hospitalizations have $\text{LOS} \leq 31$ days) and 30 days of readmission window.

We make three other sample exclusions. First, in 2015 and 2017 more conditions were added to the list of those targeted by HRRP. These conditions include chronic obstructive pulmonary disease, elective total hip arthroplasty, elective total knee arthroplasty, and coronary artery bypass graft. Even though these changes occurred after the conclusion of our observation period (2011–2014), we drop these conditions from our analysis because anticipatory behaviors by hospitals could affect our estimates. Second, the HRRP penalties only apply to hospitalizations of Medicare patients aged 65 and above. About 20% of Medicare patients are younger than 65 but have disabilities or other conditions (e.g., end-stage renal disease) that make them Medicare-eligible prior to age 65. We exclude this subgroup because prior work shows that this cohort is clinically unique, especially in terms of readmission rates (Barrett et al. 2015). Lastly, in the beginning of 2014, the ACA's Medicaid expansion extended Medicaid insurance coverage to those who were previously uninsured. To prevent potential contemporaneous effects of Medicaid coverage expansion on nontarget patient cohorts, we drop uninsured and Medicaid hospitalizations from our analysis. Additionally, for consistency, we follow prior work (Carey and Lin 2015, Mellor et al. 2017) and include only hospitalizations with primary payer of private insurance in our nontarget insurance cohorts, excluding hospitalizations with primary payer listed as other or missing. We present the details of sample definition in Figure A1 in the Online Appendix. Table 1 provides average 30-day readmission rates for all hospitalizations in the final sample.

4. Identification Strategy

4.1. Econometric Specification

Our identification strategy uses DD models to examine differential treatments along two dimensions separately: insurance and condition. This type of empirical strategy has been used previously to evaluate policy interventions (e.g., Athey and Imbens 2017, Niewoehner and Staats 2022). The DD model requires defining “pre” and “post” policy periods. HRRP penalties first went into effect in fiscal year 2013 (October 2012 to

Table 1. Average 30-Day Readmission Rate of Hospital Admissions

	(1)	(2)	(3)	(4)	(5)
Year	Cohort A	Cohort B	Cohort C	Cohort D	Overall
2011	0.199	0.154	0.112	0.069	0.118
2012	0.194	0.152	0.110	0.070	0.117
2013	0.184	0.148	0.108	0.068	0.114
2014	0.183	0.149	0.106	0.068	0.115
Pre-HRRP (Jan 2011 to Sep 2012)	0.197	0.153	0.111	0.070	0.118
Post-HRRP (Oct 2012 to Oct 2014)	0.184	0.149	0.107	0.068	0.115
Observations	1,831,271	12,188,092	504,616	11,764,809	26,288,788

September 2013). Therefore, we treat January 2011 through September 2012 as the prepolicy period and October 2012 through October 2014 as the postpolicy period (as mentioned, we exclude November and December of each year due to inadequate follow-up visibility). We note that HRRP had been in discussion and development for at least two years before its implementation. Therefore, to the extent that hospitals made improvements in anticipation of HRRP, our estimates are likely conservative estimates of the total program impact.

The DD model also requires defining a control group. HRRP affected all hospitals in the nation simultaneously, with the exception of Maryland, in which the Admission-Readmission Revenue Program was enacted in 2012 with a similar purpose as HRRP (Health Services Cost Review Commission 2011). Thus, there is not a clear “untreated” group to use as a control group. Some studies have divided hospitals into high and low performers according to their baseline readmission rates and argued that high performers can serve as a control group because they already perform well enough to avoid HRRP penalties (Mellor et al. 2017, Chen and Grabowski 2019). Although this may be true, it does not guarantee that such hospitals do not undertake readmission-reduction efforts. For example, because HRRP penalties are based on performance relative to a peer-based benchmark that is adjusted each year, even high-performing hospitals may undertake improvement efforts to avoid being penalized in future years. Another possibility is that hospitals that are part of a multihospital organization might undertake improvement efforts jointly regardless of the relative performance of any single hospital in the group.

Instead, in this study, we define the control group based on the theory that knowledge transfer and quality spillover are less likely when the source and recipient of knowledge have fewer common characteristics (Argote and Fahrenkopf 2016). This is similar to the argument in Deore et al. (2024) to define a control group in the absence of a clear untreated group. Quality knowledge is originally developed to improve quality for target patients (Cohort A). Thus, for all hypotheses, we use hospitalizations with both nontarget conditions and nontarget insurance (Cohort D) as the control group for the three other cohorts. Although there might also be quality spillover to hospitalizations in Cohort D, we choose this group because it shares no HRRP criterion with the target patients, and thus is the least likely patient group to be affected by HRRP. This approach allows us to identify the impact of HRRP on Cohorts A, B, and C relative to Cohort D, but we cannot measure HRRP-related improvements that affect all four cohorts equally. For example, a hospital might invest in information systems improvements that

benefit all patients regardless of insurance or condition. Such an improvement would be captured by the time-trend variable and would be indistinguishable from any exogenous time trends. As we discuss in Section 9, however, our results suggest that this is not a concern because the estimated time trend coefficient is almost identical to the expected exogenous time trend.

Specifically, we estimate the following linear regression model:

$$y_i = \beta_0 + \beta_1 TCohort_i + \beta_2 Post_i + \gamma_1 TCohort_i \times Post_i + \mathbf{X}_i\boldsymbol{\theta} + Condition_i + \epsilon_i, \quad (1)$$

in which y_i is an indicator of readmission occurring within 30 days of discharge for hospitalization i . The model is estimated based on a vector of covariates and whether the hospitalization occurs pre- or post-HRRP. The variable $TCohort_i$ is a binary variable indicating whether the hospitalization belongs to the treated cohort of interest (Cohort A for the main effect and Cohorts B and C for spillover effect across condition and insurance, respectively). The variable $Post_i$ is a binary variable equal to one if the hospitalization occurred in October 2012 or later.

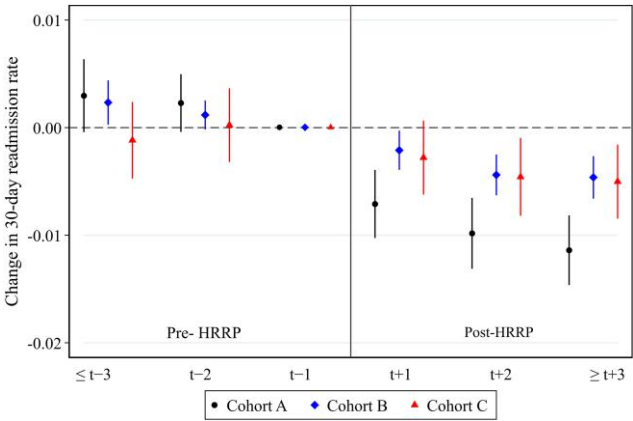
The vector $\mathbf{X}_i$ includes control variables for the patient and hospital associated with hospitalization i . These variables include the patient's age and gender, along with socioeconomic variables related to the patient's community of residence (e.g., population, median income). They also include health information related to the hospitalization (e.g., severity of illness, risk of mortality). Finally, this vector includes information about the hospital (volume, ownership, teaching status) and timing (month, the day of the week) of hospitalization discharge. The model also includes fixed effects for each admission primary condition. Table A1 in the Online Appendix provides a description of each control variable. In estimation, we cluster the standard errors at the hospital-year level to account for potential correlations in the outcome among hospitalizations occurring in the same hospital in a given year (Bertrand et al. 2004).

We choose to estimate a linear probability model rather than a binary outcome model for ease of interpretability of the coefficients. This is a reasonable choice when the emphasis of analysis is on marginal effects rather than on predictions (Wooldridge 2010). For robustness, however, we also estimate the equivalent probit and logit models in Section 8 and find that our results are robust to these specifications.

4.2. Parallel Trends Assumption

A central assumption underlying a causal interpretation of the DD econometric model is that, in the absence of treatment, treated and control groups would have parallel trends before and after the implementation of HRRP. To verify the validity of the parallel

Figure 2. (Color online) Changes in 30-Day Readmission Rate Around the Implementation of HRRP (in Event Time)



Notes. The estimates are relative to 30-day readmission rate in $t - 1$. $\leq t - 3$: Jan-Jul 2011; $t - 2$: Aug-Oct 2011, Jan-Mar 2012; $t - 1$: Apr-Sep 2012; $t + 1$: Oct 2012, Jan-May 2013; $t + 2$: Jun-Oct 2013, Jan 2014; $\geq t + 3$: Feb-Oct 2014. Bars indicate 95% confidence interval.

trends assumption, we follow the methodology in Christensen et al. (2020) and estimate the counterfactual treatment effects in six-month periods before and after HRRP. Specifically, we estimate Equation (1) separately for Cohorts A, B, and C as the treatment group and Cohort D as the control group for five periods relative to the benchmark period of April to September 2012.

We plot the estimated counterfactual treatment effects in Figure 2, which corroborates a causal interpretation of

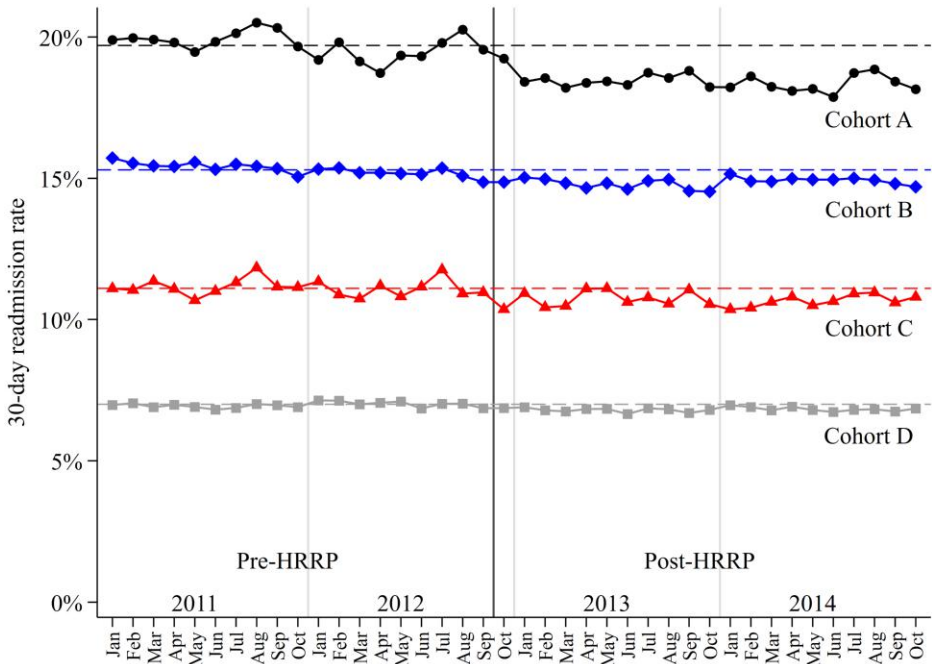
our model. This figure indicates that the effects are close to zero during the pre-HRRP period, suggesting that the parallel trends assumption is not violated. Right after the implementation of HRRP, we observe a decreasing trend in readmission rates for all three treatment cohorts relative to the control cohort, implying a causal effect.

5. Spillover Effects: Aggregate Results

We begin by examining a plot of mean levels of 30-day readmissions. Figure 3 shows the mean value of 30-day readmissions by month for each cohort and provides several immediate insights (Table 1 includes the same information by year). First, confirming the results in Section 4.2, we see that in the pre-HRRP time period (2011 to September 2012), there is no obvious departure from the parallel trends assumption. Further, we see that Cohort D exhibits no time trend and little to no noticeable change once HRRP is implemented in October 2012. This suggests (but does not prove) that HRRP had little effect on Cohort D, thus supporting its usefulness as a control group. Lastly, similar to the results in Figure 2, we note that Cohort A exhibits a marked decrease after the implementation of HRRP. Cohorts B and C appear to show some reduction as well, but the effect is smaller. These visual clues are suggestive of what we can expect from the model estimates.

We now examine the effects of HRRP on 30-day readmissions of target patients and nontarget patients

Figure 3. (Color online) Thirty-Day Readmissions Trend by Cohort



Notes. Dashed lines show average value for each cohort in 2011–September 2012. Cohort A: Medicare, target conditions; Cohort B: Medicare, nontarget conditions; Cohort C: private insurance, target conditions; Cohort D: private insurance, nontarget conditions.

Table 2. 30-Day Readmissions Change After HRRP Implementation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Main	Spillover across		Main	Spillover across		
		Condition	Insurance		Condition	Insurance	Condition
<i>TCohort</i>	0.0307*** (0.0012)	0.0158*** (0.0007)	−0.0007 (0.0012)	0.0344*** (0.0012)	0.0151*** (0.0007)	−0.0006 (0.0012)	0.0500*** (0.0030)
<i>Post</i>	−0.0019*** (0.0005)	−0.0019*** (0.0005)	−0.0019*** (0.0005)	−0.0017*** (0.0005)	−0.0016** (0.0005)	−0.0017*** (0.0005)	−0.0017*** (0.0005)
<i>TCohort × Post</i>	−0.0116*** (0.0010)	−0.0052*** (0.0007)	−0.0037*** (0.0010)	−0.0118*** (0.0010)	−0.0053*** (0.0007)	−0.0040*** (0.0011)	−0.0057*** (0.0007)
Treatment cohort	A	B	C	A	B	C	B _{sim}
Control cohort	D	D	D	D _{dis}	D _{dis}	D _{dis}	D _{dis}
Observations	13,366,960	23,538,191	12,060,288	11,949,801	22,121,032	10,643,129	13,561,375

Notes. Estimates are from linear regression models. All models include all controls in Equation (1), including condition fixed effects. Robust standard errors clustered by hospital-year in parentheses.

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

(Hypothesis 1) by estimating the DD model of Equation (1). The results are shown in columns (1) through (3) of Table 2 (Table A2 in the Online Appendix provides a full version of these columns of Table 2 with all the estimated coefficients). First, in column (1) we replicate prior analysis on the main effect of HRRP, and we see that the 30-day readmission rate for Cohort A dropped by 1.2 percentage points after HRRP implementation ($\gamma_1 = -0.0116$, $p < 0.001$). Stated as a percentage reduction, HRRP led to a 6% reduction in the 30-day readmission rate in Cohort A from the baseline rate in 2011 ($1.2\% \div 19.9\% = 6.0\%$). This result is similar in magnitude to the results reported in Wasfy et al. (2017) and somewhat smaller than the results reported in papers such as Carey and Lin (2016) and Zuckerman et al. (2016).

Hypothesis 1 posits that HRRP leads to a reduction in the 30-day readmission rate for Cohort B (spillover across condition) and Cohort C (spillover across insurance). Beginning with the spillover effects across condition, we note that the *TCohort × Post* coefficient in column (2) of Table 2 is -0.0052 ($p < 0.001$) indicating that HRRP led to a 0.5 percentage point reduction in the 30-day readmission rate for Medicare hospitalizations with nontarget conditions (Cohort B). Given the baseline 30-day readmission rate of 15.4% for Cohort B, HRRP led to a 3.2% reduction in the readmission rate ($0.5\% \div 15.4\% = 3.2\%$). Thus, we find support for Hypothesis 1 and note that the negative sign indicates that positive (beneficial) spillovers are occurring.

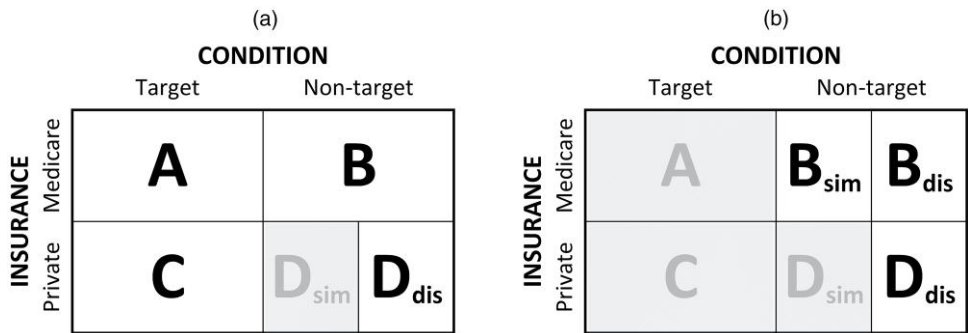
Turning to the spillover effects across insurance, we see that the *TCohort × Post* coefficient in column (3) of Table 2 is -0.0037 ($p < 0.001$), indicating that HRRP led to a 0.4 percentage point reduction in the 30-day readmission rate for hospitalizations with target conditions but private insurance (Cohort C). Given the baseline 30-day readmission rate of 11.2% for Cohort C, HRRP led to a 3.6% reduction in the readmission rate ($0.4\% \div$

$11.2\% = 3.6\%$). Thus, we find support for Hypothesis 1 and note that positive (beneficial) spillovers are occurring across insurance, just as they are across condition.

Comparing the magnitude of the main effect of the policy to the magnitude of the spillover effects, as expected, we find a larger impact on Cohort A compared with spillover Cohorts B and C. This difference is both in absolute terms ($0.4\% - 0.5\%$ for Cohorts B and C compared with 1.2% for Cohort A) and relative terms ($3.2\% - 3.6\%$ for Cohorts B and C compared with 6.0% for Cohort A). Together, these results show that the activities that lead to reduced readmission for target patients are at least partially complementary with those that reduce readmission for nontarget patients, leading to positive but limited spillover across condition and insurance.

As discussed in Section 4.1, our primary econometric specification utilizes hospitalizations with nontarget conditions and nontarget insurance (Cohort D) as the control group because this is the least likely patient group to be affected by HRRP (also supported by Cohort D data in Figure 3). There may be a concern, however, that some hospitalizations in Cohort D could still be affected by HRRP, especially if their clinical condition is similar to the targeted clinical conditions governed by the policy. To address this concern, we consider a limited version of Cohort D, restricted to hospitalizations with dissimilar conditions to the target conditions (Figure 4(a)). Specifically, to classify conditions as similar or dissimilar, we use the multilevel clinical classification software (CCS) codes. This hierarchical classification system, developed by the AHRQ, collapses more than 14,000 ICD-9-CM diagnosis codes into a smaller number of clinically meaningful nested categories (Elixhauser et al. 2015). HRRP's target conditions AMI, HF, and PN have CCS codes 7.2.3, 7.2.11, and 8.1.1, respectively. AMI and HF are both nested under category 7, Diseases of the Circulatory System. PN is

Figure 4. Patient Cohorts with Nontarget Conditions Similarity Consideration



nested under category 8, Diseases of the Respiratory System. Thus, Cohort D_{dis} includes hospitalizations with conditions that are neither target conditions nor similar to any target conditions (i.e., all CCS codes other than 7.x.y and 8.x.y). We repeat the analyses in columns (1)–(3) of Table 2 with D_{dis} as the control group and present the results in columns (4)–(6) of Table 2. A pairwise comparison of the *TCohort* × *Post* coefficient in columns (4)–(6) and columns (1)–(3) shows that the estimates of the spillover effects, as well as the main effect, are not sensitive to the control group selection (−0.0118 versus −0.0116, −0.0053 versus −0.0052, and −0.0040 versus −0.0037), indicating that Cohort D is an equally valid control group.

Finally, along the same lines as the above discussion, one might argue that spillover across condition should only be measured for clinically similar conditions to the targeted ones, as these conditions better represent the hospitalizations that may be indirectly affected by the policy. Therefore, we restrict hospitalizations in Cohort B to those with a similar condition to AMI, HF, and PN (Figure 4(b)). Specifically, Cohort B_{sim} includes hospitalizations with circulatory conditions other than AMI and HF (i.e., CCS codes 7.x.y, excluding 7.2.3 and 7.2.11), as well as respiratory conditions other than PN (i.e., CCS codes 8.x.y, excluding 8.1.1). In column (7) of Table 2, we repeat the analysis from column (5) but with only B_{sim} as the treatment group rather than all of Cohort B. The coefficients of the interaction term indicate no significant change in the estimates of the spillover effect across condition (−0.0057 versus −0.0053). Note that in Section 6.1, we formally test the hypothesis that clinical condition similarity moderates the spillover effect across condition.

6. Spillover Moderators

Having shown that spillover occurs across condition and insurance for 30-day readmissions (Hypothesis 1), we now present additional analyses to gain a richer understanding of these results and provide answers to the second and third research questions posed in Section 1. Our goal is to explore whether there is

heterogeneity in spillover effects with respect to two important features: (1) clinical condition similarity and (2) hospital operational focus.

6.1. Task-Level Moderator: Clinical Condition Similarity

We now consider if the spillover effect across condition is stronger for some nontarget conditions than others. Hypothesis 2 states that HRRP leads to a relatively larger reduction in 30-day readmissions for nontarget conditions that are similar to target conditions.

To test this, we use the model in Equation (1) but adjust the definition of the four subgroups used in this model (Figure 4(b)). Because our focus is now on the effect of HRRP on hospitalizations with similar and dissimilar *nontarget* conditions, we exclude hospitalizations with target conditions from the estimation (Cohorts A and C), and as explained in Section 5, we split Cohorts B and D into nontarget conditions that are clinically similar or dissimilar to the target conditions. If HRRP had no spillover beyond Cohort A, then there would be no treatment effect of HRRP on *any* of the nontarget cohorts in Figure 4(b). However, given that there is spillover to nontarget conditions (Hypothesis 1, Section 5), we test if the spillover to Cohort B_{sim} is of greater magnitude than the spillover to Cohort B_{dis}.

Table 3 presents the results of three separate estimations of Equation (1). In column (1), Cohort B_{sim} includes hospitalizations with circulatory conditions other than AMI and HF (i.e., CCS codes 7.x.y, excluding 7.2.3 and 7.2.11), and in column (2), Cohort B_{sim} includes hospitalizations with respiratory conditions other than PN (i.e., CCS codes 8.x.y, excluding 8.1.1). For all models, the dissimilar cohorts (Cohorts B_{dis} and D_{dis}) include hospitalizations with conditions that are neither target conditions nor similar to any target conditions (i.e., all CCS codes other than 7.x.y and 8.x.y), and we use D_{dis} as the control group.

In columns (1) and (2) of Table 3, we find spillover effects of HRRP on both sets of similar conditions ($\gamma_1 = -0.0047$, $p < 0.001$ for circulatory conditions and $\gamma_1 = -0.0098$, $p < 0.001$ for respiratory conditions). In

Table 3. Heterogeneity of Spillover Effects Across Condition (with Respect to Clinical Condition Similarity)

	(1)	(2)	(3)
	Similar to		Dissimilar to target conditions
	AMI & HF	PN	
<i>TCohort</i>	0.0451*** (0.0034)	0.0513*** (0.0031)	0.0157*** (0.0007)
<i>Post</i>	−0.0016*** (0.0005)	−0.0017*** (0.0005)	−0.0016** (0.0005)
<i>TCohort × Post</i>	−0.0047*** (0.0007)	−0.0098*** (0.0013)	−0.0052*** (0.0007)
Treatment cohort	B_{sim}	B_{sim}	B_{dis}
Control cohort	D_{dis}	D_{dis}	D_{dis}
Observations	12,868,072	10,841,837	18,708,191

Notes. Estimates are from linear regression models. All models include all controls in Equation (1), including condition fixed effects. Robust standard errors clustered by hospital-year in parentheses.

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

column (3), we also find spillover to conditions that are clinically dissimilar (i.e., noncirculatory and nonrespiratory conditions) ($\gamma_1 = -0.0052$, $p < 0.001$). To test Hypothesis 2, we compare the effects on similar conditions (columns (1) and (2)) to the spillover effect on dissimilar conditions (column (3)). Figure 5 depicts the estimated interaction coefficients and 95% confidence intervals. These comparisons indicate mixed support for Hypothesis 2. The spillover effect on similar respiratory hospitalizations is significantly larger than the reduction for dissimilar hospitalizations (difference between γ_1 in columns (2) and (3) = -0.0046 , $p < 0.001$). The magnitudes of the spillover effect on similar circulatory hospitalizations, however, is not statistically different than the

spillover effect on dissimilar conditions ($p > 0.1$). One possible explanation for these different results for AMI-HF and PN could be that hospitals have employed different interventions to reduce readmissions for various target conditions, which might impact similar clinical conditions differently. An opportunity for future research would be to delve more deeply into the condition-specific interventions and how they impact quality improvement spillovers and their moderators.

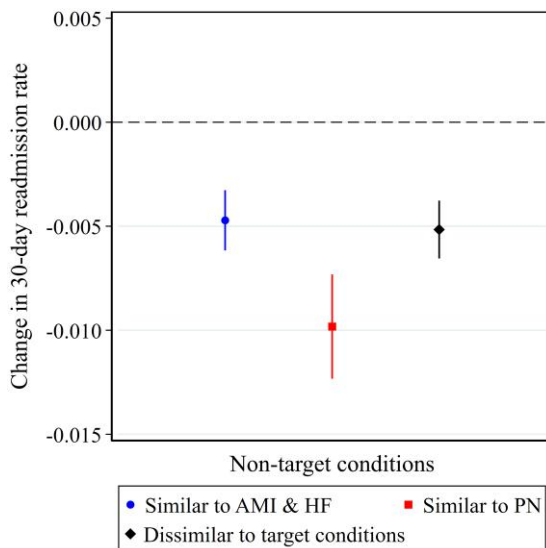
6.2. Organizational-Level Moderator: Hospital Operational Focus

Hypothesis 3 states that HRRP leads to a larger reduction in 30-day readmission rate of Cohorts B and C in hospitals with higher operational focus on target patients. To test this hypothesis, we classify hospitals by their Cohort A proportion. Specifically, we define focus for a given year in hospital i as the number of Cohort A hospitalizations in hospital i (n_{iA}) during that year, divided by the total number of hospitalizations in hospital i during that year,

$$Focus_i = \frac{n_{iA}}{\sum_{j=A,B,C,D} n_{ij}}, \quad (2)$$

where n_{ij} is the total number of hospitalizations in cohort j in hospital i in a given year. Note that we use the original cohort definitions depicted in Figure 1.

Following Kuntz et al. (2019), we categorize hospitals as having high or low focus by splitting at the median value of $Focus_i$ (8.8%). However, because $Focus_i$ is strongly (negatively) correlated with hospital volume (discharges per year), to rule out potential confounding factors associated with hospital volume and to avoid complications of numerous interaction terms, we estimate Equation (1) separately for each hospital volume-focus level. For volume, we employ its 33rd and 67th

Figure 5. (Color online) Heterogeneity of Spillover Effects Across Condition (with Respect to Clinical Condition Similarity)

Note. Bars indicate 95% confidence interval.

Table 4. Heterogeneity of Spillover Effects (with Respect to Hospital Operational Focus)

	(1)	(2)	(3)	(4)	(5)	(6)
	Small		Medium		Large	
	Low	High	Low	High	Low	High
<i>TCohort</i>	0.0030 (0.0046)	0.0135*** (0.0027)	0.0148*** (0.0015)	0.0108*** (0.0020)	0.0136*** (0.0009)	0.0196*** (0.0017)
<i>Post</i>	−0.0020 (0.0030)	−0.0012 (0.0015)	−0.0019* (0.0010)	−0.0030** (0.0010)	−0.0017** (0.0006)	−0.0027* (0.0011)
<i>TCohort × Post</i>	−0.0067† (0.0040)	−0.0088*** (0.0022)	−0.0071*** (0.0015)	−0.0047*** (0.0014)	−0.0045*** (0.0009)	−0.0033* (0.0016)
Treatment cohort	B and C	B and C	B and C	B and C	B and C	B and C
Control cohort	D	D	D	D	D	D
Observations	159,142	546,642	2,926,393	1,935,112	15,747,525	2,717,972

Notes. Estimates are from linear regression models. All models include all controls in Equation (1), including condition and hospital fixed effects but excluding hospital volume. Robust standard errors clustered by hospital in parentheses.

† $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

percentile values to categorize hospitals into small, medium, and large categories.

Table 4 presents the results of these estimations. We see that all the *TCohort × Post* coefficients in columns (1) through (6) are negative and statistically significant, indicating that there is positive quality spillovers to non-target visits for all levels of focus across all hospital volumes. The magnitude of readmission reduction ranges from 0.3 to 0.9 percentage points. To test whether the effect sizes are significantly different between focus levels within a given hospital volume level, we conduct pairwise comparison tests between the coefficients of the interaction term in Table 4 and find no significant difference in spillover by focus level. Figure 6 depicts the estimated interaction coefficients and 95% confidence intervals, which exhibit a great deal of overlap. Thus, we find that the spillover effects of HRRP are not moderated by operational focus in hospitals, and we do not find support for Hypothesis 3. Recall that even hospitals with a low operational focus on target patients

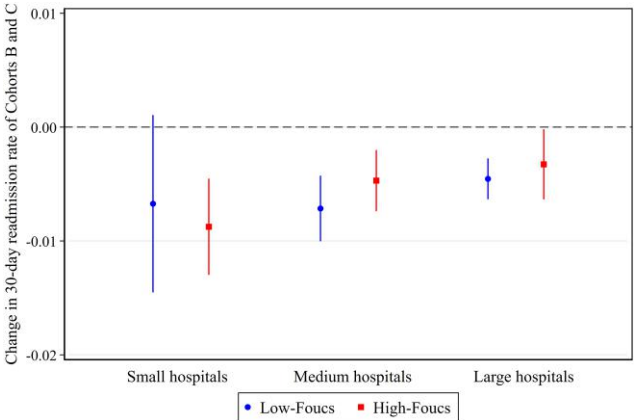
may still have sufficient incentives to reduce readmissions for these patients, as HRRP’s penalties are applied to total Medicare reimbursement: not just the reimbursement associated with target hospitalizations.

7. Post Hoc Analysis: Care Intensity in Hospitals

As mentioned in Section 2, hospitals can implement various interventions to reduce readmissions. These interventions include holding patients longer before discharge (Anderson et al. 2012, Krishnan and Ramasubramanian 2023), providing more intense and thorough care during the visit (Carey 2015), or through coordinating postdischarge care, such as more comprehensive discharge instructions or proactive scheduling of follow-up care (Brewster et al. 2016). Note that the first two interventions likely lead to increased cost per hospitalization, whereas the cost implications of the third intervention are ambiguous. We now look for evidence of what interventions hospitals employed for target patients and whether these interventions spilled over to nontarget patients.

We examine three common measures of hospital care intensity for evidence of change under HRRP: length of stay, procedure count, and cost (Wennberg et al. 2009, Antwi et al. 2015, Castaneda and Saygili 2016). For this analysis, we focus on hospitalizations not preceded by another hospitalization within the prior 30 days, including only those that are not readmissions. This approach allows us to measure care intensity independent of services provided during a potentially related prior visit. Because individual patients cannot be tracked from one year to the next, a hospitalization in the first two months of each year might possibly be a 30-day readmission for a prior hospitalization. Thus, we exclude hospitalizations occurring in the first two months of each year to account for one month of possible hospital stay and 30 days of

Figure 6. (Color online) Heterogeneity of Spillover Effects (with Respect to Hospital Operational Focus)



Note. Bars indicate 95% confidence interval.

readmission window. Table A3 and Figure A2 in the Online Appendix provide summary statistics of the care intensity measures for the NRD.

We estimate Equation (1) using three dependent variables: LOS, number of procedures, and cost. LOS is measured by the count of midnights encompassed by the hospitalization, and the number of procedures is determined by the count of ICD-9-CM procedures coded on the discharge record. The cost variable warrants further explanation. The NRD includes “charges” as reported by the hospital, which is a function of many factors including the clinical condition, the procedures performed, and the LOS (Philbin et al. 2001, Polverejan et al. 2003, Reinhardt 2006). However, charges vary greatly across hospitals for the same services and, with lack of price transparency, often do not reflect the actual amount paid by payers (Jiang et al. 2023). To address this, the Centers for Medicare & Medicaid Services (CMS) estimates a “cost to charge ratio” (CCR) for each hospital each year, enabling a more accurate comparison of costs across hospitals (Healthcare Cost and Utilization Project 2018). Following common practice in the literature, we use these hospital-year CCRs to convert charges to costs, which we use as a measure of care intensity (Chen et al. 2010, Senot et al. 2015). Finally, we convert each hospitalization’s cost to 2011 U.S. dollars using the consumer price index for inpatient hospital services (Bureau of Labor Statistics 2012, 2013, 2014).

The estimation results are shown in Table 5. In columns (1)–(3), the $TCohort \times Post$ coefficients are all not significant, indicating that LOS did not change for target patients or for either of the spillover cohorts. Thus, we do not find evidence that readmission reductions were achieved through holding patients in the hospital longer.

For the number of procedures (columns (4)–(6)), we see only marginally significant evidence of a very small increase in the mean number of procedures per hospitalization for target patients (0.0287, $p < 0.10$). We do not

observe a significant change for either of the spillover cohorts. Thus, we find no evidence that readmission-reduction spillover occurred through an increase in procedures for spillover cohorts.

For cost per hospitalization, we do not find a significant change in cost for target patients or for Medicare patients with nontarget conditions (Cohort B). We do, however, find significant cost reductions for hospitalizations with target conditions and private insurance (Cohort C). The total effect on cost for Cohort C is a \$502 reduction ($p < 0.001$), which is a 3.3% reduction from the 2011 baseline average cost of \$15,060. This cost reduction, however, is likely not a result of quality improvement spillovers because Cohort A did not experience a similar cost reduction. Future research could explore the sources of these cost reductions and why they are limited to privately insured patients with target conditions.

The fact that neither LOS, procedure count, or cost increased as readmissions decreased suggests that hospitals implemented cost-effective changes in care delivery that created real improvements for patients rather than initiating costly, resource-intensive interventions to avoid HRRP penalties.¹ The similarity in results across cohorts in Table 5 suggests that the spilled over readmission reductions for Cohorts B and C are caused by the same types of actions taken to reduce readmissions for Cohort A, as one would expect.

Prior literature has documented various readmission-reduction interventions used by hospitals, such as fostering collaboration across hospital departments (Brewster et al. 2016), improving adherence to medication guidelines (Andritsos and Tang 2014), enhancing patient education for postdischarge care and medication (Bradley et al. 2012), proactively arranging follow-up appointments and home visits (Kripalani et al. 2014), and expanding the use of information technology support systems to track readmissions and share information with inpatient and outpatient care providers (Figueroa

Table 5. Measures of Care Intensity Change After HRRP Implementation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	LOS			Number of procedures			Cost		
<i>TCohort</i>	−0.3820*** (0.0403)	−0.1352*** (0.0218)	0.4049*** (0.0394)	−0.3240*** (0.0187)	−0.0679*** (0.0121)	−0.1827*** (0.0177)	−640.09*** (133.83)	−730.90*** (72.44)	1,825.99*** (158.50)
<i>Post</i>	−0.0712*** (0.0186)	−0.0635*** (0.0186)	−0.0711*** (0.0186)	−0.0057 (0.0219)	−0.0040 (0.0219)	−0.0056 (0.0220)	−397.95*** (110.70)	−356.11** (112.09)	−396.76*** (111.14)
<i>TCohort × Post</i>	−0.0024 (0.0263)	−0.0179 (0.0218)	−0.0061 (0.0312)	0.0287† (0.0171)	−0.0031 (0.0116)	0.0037 (0.0220)	−158.40 (99.42)	−61.84 (66.54)	−502.48*** (137.22)
Treatment cohort	A	B	C	A	B	C	A	B	C
Control cohort	D	D	D	D	D	D	D	D	D
Observations	9,725,206	16,663,341	8,940,012	9,725,272	16,663,543	8,940,071	9,278,690	16,015,421	8,509,581

Notes. Estimates are from linear regression models. All models include all controls in Equation (1), including condition fixed effects. Data are limited to first visits in a 30-day time window. Robust standard errors clustered by hospital-year in parentheses.
† $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

et al. 2017). For comprehensive surveys of the literature on readmission-reduction interventions, we refer the reader to Hansen et al. (2011), Brewster et al. (2016), and Wahl and Hawn (2017). Although this literature does not directly link the use of any of these interventions to HRRP, it is likely that many of these interventions could result in the observed spillovers. For example, if a department creates an improved patient discharge education process, it is likely that the process will be used for all patients in the department, regardless of insurance type or specific condition. Similarly, increased cross-department communication presumably benefits many patients regardless of insurance or condition. An interesting avenue for future research would be to survey hospitals to learn what interventions they have used to achieve readmission reductions in response to HRRP.

8. Robustness Checks

We conduct a variety of robustness checks to verify that our results are not sensitive to model specifications.

8.1. Probit and Logit Specifications

The 30-day readmission metric is a binary dependent variable, and our main estimates are from linear probability models. To ensure that our results are not driven by the linear specification, we repeat the analyses in Table 2 using probit and logit models. The results of these analyses are shown in Table A4 in the Online Appendix. To make these results comparable to our earlier estimates, we present the marginal effects of HRRP on different cohorts implied by these model coefficients. We observe that the results are directionally unchanged and similar in magnitude and statistical significance to our earlier results.

8.2. Difference-in-Difference-in-Differences (DDD) Specification

Our results are based on the DD identification strategy. However, by construction, three DD models can be embedded in a single DDD model. For example, all three DD models in columns (1) through (3) of Table 2 can be estimated in a single DDD model. In Table A5 in the Online Appendix, we use a DDD model to estimate the relative change in the target metric for these cohorts in a single model. As expected, we find similar results in magnitude and statistical significance to those from the DD models in the corresponding columns of Table 2, confirming HRRP spillover effects across insurance and across condition, as well as readmission improvements for Cohort A. Note that in a typical DDD analysis, one would normally examine the $Medicare \times Target \times Post$ interaction term to check whether there is a reduction in the 30-day readmission rate for Cohort A. We see that here, it is not statistically significant, and thus might be tempted to conclude that HRRP has no

effect on Cohort A, the target hospitalizations. However, because we show that spillover occurs across condition and insurance, the coefficient of the triple interaction term does not measure the actual total treatment effect on target Cohort A; it only shows the effect of HRRP on Cohort A less any spillover effects that the policy has. Instead, the total effect of HRRP on Cohort A relative to Cohort D is captured by the sum of the coefficients of three interaction terms: $Medicare \times Post$, $Target \times Post$, and $Medicare \times Target \times Post$ (-0.0113 , $p < 0.001$).

8.3. Hospital-Year Fixed Effects

Because of the NRD's annual change in disguised hospital identifiers, we cannot track hospitals across years and thus are unable to include hospital fixed effects in our main analysis. Instead, we control for hospital covariates such as volume, ownership, and teaching status. There may be a concern, however, that these covariates do not fully capture all unobservable hospital-related factors. To address this concern, we re-estimate our analyses of Table 2 by including *hospital-year* fixed effects (our preferred estimates include the covariates instead of the hospital-year fixed effects as their interpretation is more transparent). The results of this new analysis are shown in Table A6 in the Online Appendix, and all columns demonstrate the stability of our earlier estimates.

8.4. Matched Conditions in Treatment and Control Cohorts

HRRP deliberately targets clinical conditions with relatively high readmission rates. One may be concerned, however, that our identified effects could be an overestimate of the true effects because conditions in Cohorts A and C are riskier and have more room for potential improvements. To address this concern regarding non-random selection of the treatment group, we repeat the analysis in columns (1) and (3) of Table 2 but include only selected clinical conditions in the control group (Cohort D). Specifically, to estimate the main effect for each target condition in Cohort A, we find one nontarget condition in Cohort D that has the closest pre-HRRP 30-day readmission rate: AMI, HF, and PN are, respectively, matched with cancer of pancreas, pathological fracture, and unspecified cancer (baseline readmission rates of 17.9% versus 18.0%, 22.9% versus 22.8%, and 16.9% versus 17.1%). To estimate the spillover effect across insurance for each target condition in Cohort C, we repeat the same process of identifying a matching nontarget condition in Cohort D: AMI, HF, and PN are, respectively, matched with digestive congenital anomalies, unspecified cancer, and urinary tract infections (baseline readmission rates of 8.2% versus 8.2%, 17.9% versus 18.0%, and 9.7% versus 9.7%). The results of these analyses are shown in Table A7 in the Online Appendix. Although not statistically different from each other, the estimates in Table A7 are larger in

magnitude compared with the estimates in Table 2 and thus do not provide any evidence for an overestimated effect in our main analysis.

8.5. Age Subgroups

HRRP targets Medicare patients and thus is primarily concerned with patients aged 65 and older, but our analysis includes younger patients in the comparison groups (Cohorts C and D). There may be a concern that these younger patients could have different readmission probabilities even after controlling for demographics, condition fixed effects, and the other controls in our model. To address this concern, we repeat the analysis in columns (1) through (3) of Table 2 on three subgroups of privately insured patients (Cohorts C and D): those aged 40 and older, 50 and older, and 60 and older. There is no change to the Medicare Cohorts A and B. The results are shown in Table A8 in the Online Appendix. We observe that the spillover effects across insurance and condition, as well as the total effect on target patients, hold within each subsample.

9. Discussion and Conclusion

In this paper, we study quality improvement spillovers of partial incentives in a multitask service setting by investigating quality spillovers in the hospital context. Specifically, we explore the effects of a national healthcare quality improvement policy, the HRRP, on 30-day readmissions for patients who are not targeted by the policy. We conduct analyses of HRRP using a national-level database of millions of hospitalizations over four years. To test for spillover effects across the two targeted attributes of patients (insurance and condition), we estimate several difference-in-differences regression models, which allow us to separately identify these two types of spillover effects.

Our findings show that HRRP led to strong quality improvement spillovers to nontarget patients. In particular, we find that quality spillovers occur both to Medicare patients with nontarget conditions (i.e., spillover across condition) and to patients with target conditions but private insurance (i.e., spillover across insurance). These results are in contradiction to the main conclusions of the principal-agent framework of Holmstrom and Milgrom (1991). We note that this discrepancy may be explained by several misalignments between the core assumptions of Holmstrom and Milgrom (1991) and the operation of the U.S. healthcare control system. First, the main analysis in Holmstrom and Milgrom (1991) assumes that different tasks are substitutes, whereas in the hospital setting, clinical care processes for different patients may be complements. Second, in contrast to the setting in Holmstrom and Milgrom (1991), quality performance metrics (e.g., readmission rate) are easily measurable in hospitals. Third, unlike the framework of

Holmstrom and Milgrom (1991), in the U.S. healthcare system, the principal (i.e., the government) does not have control over assigning specific tasks exclusively to specific agents (i.e., hospitals). Fourth, in the hospital setting, many tasks are performed simultaneously, which contrasts with the explicit temporal assumption made in Holmstrom and Milgrom (1991).

Further, our results indicate that patients with conditions similar to target conditions benefit more from quality improvement spillovers than do patients with conditions dissimilar to target conditions. These findings suggest task similarity as a novel moderator for quality improvement spillovers of partial incentives in multitask settings. This effect is likely due to increased task complementarity between patients with similar conditions. In addition, we do not find evidence of spillover being moderated by the proportion of a hospital's visits that are from target patients (i.e., focus on Cohort A). Altogether, these findings contribute to the growing stream of research on partial incentives in multitask settings, examining moderators of spillovers: temporal proximity (Hecht et al. 2012), task complementarity (Jung and Shin 2022), and across-agent cooperation necessity (Gallani 2023).

In our post hoc analysis, we find no significant change in the care intensity measures of LOS and the number of procedures, and either no change or a decrease in cost per hospitalization. These results are important because to reduce readmissions, rather than cost-effective changes, hospitals could implement resource-intensive interventions. On the contrary, our findings suggest that hospitals undertook process improvement initiatives in response to HRRP with beneficial effects well beyond the narrow focus of this legislation.

9.1. Managerial and Policy Implications

Although the estimates in this paper are specific to HRRP, our findings more generally demonstrate that it is possible to utilize a quality improvement policy with a narrow target to achieve broader results in a multitask setting by exploiting beneficial quality improvement spillovers. This is particularly useful when the cost of administering a policy grows with its complexity, be it a public policy or an internal corporate policy. Policymakers should thus consider and evaluate the potential spillover effects stemming from such policies. To fully exploit this possibility, it is important to be mindful of the conditions under which quality improvement spills over. Specifically, an immediate implication of our finding on task similarity is that to achieve a broad impact with a narrow policy, the policy should target dissimilar tasks (e.g., clinical conditions). This is because quality knowledge spills over more easily to similar tasks, thus generating a broader impact.

Although the focus of this study is on regulatory reimbursement policy in healthcare, it is reasonable to

expect that the implications of our findings extend beyond this setting. For example, in the context of food safety regulations, the European Food Safety Authority (EFSA) recently banned titanium dioxide, a common additive in confectionaries, due to potential genotoxicity (European Commission–Health and Food Safety 2022). In contrast, the Food and Drug Administration (FDA) and Health Canada have not imposed similar bans. Manufacturers that produce the same products for both European and North American markets, however, might reformulate their products for both markets, resulting in positive spillovers for North American consumers. These expected positive spillovers are comparable to those across insurance in this paper. An immediate implication of this type of spillover is that, in its presence, some principals (FDA and Health Canada in this case) might be able to “free-ride” on regulations enacted by other principals (EFSA in this case). Similarly, in auto safety regulations, standards aimed at improving frontal crash safety could lead manufacturers to adopt designs that also enhance overall vehicle rigidity and safety in various crash scenarios not specifically targeted by the regulation. These positive spillovers are comparable to those across condition in this paper. If such positive spillovers exist, they could reduce the need for multiple specific regulations in auto safety. Regulators, however, must ensure these spillovers occur before relying on narrow regulations.

In addition to regulations in the public sector, our results speak to private firms with comparable organizational structures. For example, the U.S. healthcare system and private conglomerate firms share similarities that suggest our findings can be generalized. As an illustration, in the automotive manufacturing setting studied by Muthulingam and Agrawal (2016), suppliers produce similar (but not identical) parts for multiple customers, with one customer providing technical assistance and encouragement for quality improvements. This multitask setting parallels our study, where CMS acts as the principal and hospitals as agents. The spillover effects observed across insurance, where CMS incentives lead to improvements for other payers, mirror the quality improvements benefiting multiple customers in the automotive context. Similarly, our analysis of spillovers across conditions, where improvements for Medicare patients with target conditions benefit those with nontarget conditions, is analogous to the spillover of quality improvements in one part for the encouraging customer to another part for the same customer.

We note that, although our study focuses on the impact of partial incentives, specifically HRRP penalties, regulations in other industries do not always involve direct financial incentives or penalties. For example, the regulatory ban of titanium dioxide in confectionaries or auto safety regulations mandate specific

standards that manufacturers are required to meet, rather than linking noncompliance to financial penalties. Nevertheless, these examples still represent instances of partial targeting. Such regulations often impose rules, fines, or mandate disclosures, affecting only specific tasks or dimensions of a firm’s operations while leaving other areas unregulated. Our findings suggest that such partial targeting, regardless of whether it involves direct incentives or other forms of regulatory actions, can lead to spillover effects in multitask settings. Thus, our study’s insights may extend beyond penalty incentives to various regulatory actions, highlighting their potential for broader quality improvements.

Finally, although this study only examined a penalty incentive program, there may be some symmetry in how rational actors, such as hospitals, respond to penalty versus bonus incentives. In a laboratory experiment, Christ et al. (2012) show that compared with partial *penalty* incentives, partial *bonus* incentives create more trust in agents, generating stronger spillover effects. Indeed, prior empirical studies document such positive spillovers in response to bonus incentives in various settings (Muralidharan and Sundararaman 2011, Jung and Shin 2022, Wang et al. 2022). In addition to the spillover magnitude, the underlying mechanisms may vary for penalty versus bonus incentives. For example, Wang et al. (2022) hypothesize reciprocity as a driver of positive spillovers, a mechanism that does not apply to penalty incentives.

9.2. Limitations

As with any study, the present analysis has some limitations. The first limitation is the availability of an ideal control group. As noted in Section 4.1, we use hospitalizations with nontarget conditions and private insurance (Cohort D) as the control group for the three other cohorts. As such, any effect of HRRP on Cohort D cannot be separately identified from the time trend coefficient (β_2). Zuckerman et al. (2016) estimate that prior to HRRP and the ACA (2007–2010), 30-day readmissions for nontarget conditions were declining by about 0.1 percentage points per year. If that underlying non-HRRP reduction was to continue into the study observation period, we would expect β_2 to be about -0.002 , as it captures the average change for two consecutive years with similar sample sizes. Columns (1) through (3) of Table 2 show an estimated β_2 of -0.0019 , suggesting that HRRP had no impact on Cohort D.

Second, a number of changes were introduced to the U.S. healthcare system toward the end of the time frame of our data. As discussed in Section 3, one significant change was the Medicaid insurance expansion under the ACA at the beginning of 2014. The ACA significantly broadened Medicaid eligibility for those previously uninsured, but this likely does not impact our results as both uninsured and Medicaid hospitalizations

are excluded from the analysis. Further, we observe similar magnitudes of the spillover effects in 2013 and 2014 (results not shown), suggesting that the Medicaid coverage expansion had limited impact on our setting. Another change was the introduction of two other CMS pay-for-performance programs: the Hospital-Acquired Condition Reduction Program (HACRP), and the hospital Value-Based Purchasing (VBP) program (Centers for Medicare & Medicaid Services 2019). HACRP was enacted in October 2014, which falls outside the observation period of our study. VBP was established at the same time as HRRP, but recent studies show that the VBP did not contribute to improved outcomes (Ryan et al. 2017, Norton et al. 2018, Gupta 2021), and thus the results in this paper are unlikely to be due to VBP.

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Endnote

¹ We also investigate the changes in readmission rate during a non-target time period by HRRP (i.e., 31–60 days) and find a significant reduction in readmission rates for Cohorts A, B, and C within this extended time frame (Table A9 in the Online Appendix). These results provide additional confirmation that hospitals implemented meaningful quality improvement interventions that extend beyond HRRP's target of 30 days. We thank an anonymous referee for this suggestion.

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